

level changes. Much like Linux or OS X handles them today, an admin-level task will require you to enter the pertinent password, or at least click on a confirmation.

Since a service under Vista will inherit the rights of its user or application, a service will therefore be unable to make those changes as well. As a practical example, Internet Explorer 7 will be able to run in a low-level mode, wherein it will be able to access all Web sites. However, it will be difficult to launch a malicious attack from this instance of IE, since IE will have no rights to system or user documents.

Notably, Internet Explorer 7 will also offer anti-phishing functionality via blacklists and sniffing of spoof URLs. Add to this a firewall that will be able to block and filter both incoming and outgoing packets, and an integrated anti-spyware engine, and Vista will offer a first line of defence against the most common attack vectors.

With the introduction of Windows Vista, Microsoft will also offer a decent palette of parental controls to the millions of Windows users. This feature will allow you to control your child's activities on a computer; it will even main-



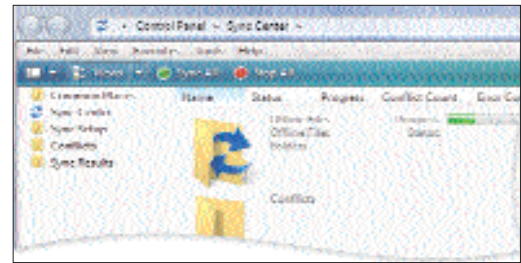
The Windows Sidebar will collate information from both Web sites as well as your desktop computer

tain a log. This can be something as simple as blocking a game based on its ESRB rating, blocking specific programs, and enforcing Web restrictions and time limits. A parent will thus be able to specify that his or her child cannot use the computer between 6 PM and 9 PM on weekdays, and further, that the child can only use it for three hours on weekends, with access to pornography blocked and with downloading disabled.

Vista will offer data encryption under specific Intel Trusted Platform Module hardware. The OS will be able to encrypt the entire hard disk, thus preventing data theft. It will also be able to deny booting with devices such as DVD drives, floppy disks, or USB drives: this will prevent malicious break-ins and avoid data theft. IT administrators can also use Group Policy in Windows Vista to block the installation of removable storage devices.

Under the operating system, security updates and system patches can happen in small increments. It can monitor network inactivity and trickle in an update, thus reducing overhead and making the most efficient use of bandwidth.

Perhaps the most interesting element of Vista is that the OS is "componentised" by design. This is roughly similar to how Linux works—a central kernel, connected to services and daemons that



The Sync Center: Vista will allow data synchronisation with both mobile devices and servers

add functionality as and when needed. Similarly, Vista will have a core operating system, called the Windows Pre-Installation Environment (WinPE), which will be language-independent and a subset of all the versions of Vista that Microsoft plans to release.

WinPE will be available as a single image on a Vista install disk. The setup routine will only need to copy this image to the system and reboot it to detect hardware—this will vastly reduce the setup time for the OS. Furthermore, you can also back up an installed system to an image, adding incremental changes to the image as and when needed.

Such an incremental means of data backup is pervasive in Vista. Notably, Windows Backup is enhanced by a new task-scheduling system that allows for status-based triggers, along with the usual timed events. For example, Vista can trigger a backup of your data if 80 per cent of your hard disk has been used up.

Interestingly, Vista will also offer a state-tracking system called Previous Versions. This will track folders, documents and such for changes, and if you desire, it will roll back the tracked element to a previous version. If you've ever opened a document, experimented with it and pressed [Ctrl] + [S] accidentally, thus losing the original, you can understand why this feature will be a boon.

Ringing In The New

In many ways, Windows Vista is a parting with old ways and ideas. Compatibility with legacy applications has been sacrificed for greater security. Bleeding-edge hardware, especially video cards and high-definition screens are being embraced. The OS is very network-aware, and the data flood that this awareness brings will hopefully be reined to control using metadata-driven storage. Mobile devices are embraced, and data portability and interoperability is enhanced.

Most importantly, like its predecessors, Vista promises to be a great platform for even more radical ideas than those dreamt at Redmond. With XAML and the .NET platform's evolution to WinFX, Vista offers a comprehensive bundle of technology that will allow a developer to quickly build applications—both online and off.

Windows Vista offers change. It will certainly demand its pound of hardware upgrades and will come with teething problems, but is perhaps the most intriguing release from Microsoft since Windows 95 first replaced the command prompt with the Start button. ■

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